FACET: Measuring Attribution Faithfulness in Multi-Factor LLM Reasoning

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Abstract

We study how the format of an attribution probe changes the picture of large language model behavior on multi-factor reasoning tasks. The standard measurement asks the model "which single factor is most important?" Here we show that this question systematically misrepresents what the model is doing. We compare two probes on the same nine frontier configurations from six labs (Anthropic, OpenAI, Meta, Mistral, DeepSeek, Alibaba/Qwen), evaluated on ten real US tort-law balancing cases and three counterfactual variants in which the canonical factor is explicitly neutralized. The single-factor probe makes all 8 models look lexicographically collapsed on a Barker-style risk-utility counterfactual: every model names the neutralized factor as most important. The weighted-rank probe, applied to the same models and cases, makes all 8 models look fully integrated: every model assigns the neutralized factor 0% weight and distributes the rest across the other factors. The contradiction is the result. An adversarial test removing the cue phrases "unresolved" and "not load-bearing" from the neutralized factor reveals a secondary split: all five open-weight models stay at 0%; the three closed frontier models (Claude Opus, Claude Sonnet, GPT-5.4) shift to 25%, 10%, and 30%. Closed models read cue phrases; open models read evidence. Prior findings of attribution unfaithfulness in multi-factor reasoning should be re-examined under weighted-rank probing, and probes for closed models need adversarial validation.

1 Introduction

1.1 Summary of Findings

Probe format determines the picture. Single-factor probes ("which one is most important?") and weighted-rank probes ("assign weights summing to 100") applied to the same 8 frontier model configurations on the same legal balancing cases produce contradictory pictures of attribution behavior. The contradiction reproduces across all three counterfactual doctrines tested.

Single-factor probes overstate lexicographic collapse. On a Barker-style risk-utility counterfactual where the canonical factor is explicitly neutralized, 0/8 models shift their attribution under the single-factor probe; 8/8 shift under the weighted-rank probe.

Frontier LLMs do not lexicographically collapse on real cases. Under the weighted-rank probe, no model assigns $\geq$50% of the weight to any single factor on any of three real anchor cases. Top factors average 22–28%; the remaining weight distributes across the other 4–6 factors.

Weighted-rank probes are partially signal-word dependent for closed models. An adversarial test that removes explicit neutrality cue phrases ("unresolved," "not load-bearing") from the neutralized factor splits the model set: all five open-weight models stay at 0%; Claude Opus shifts to 25%, Claude Sonnet to 10%, GPT-5.4 to 30%. Closed models read cue phrases; open models read evidence.

Inference-time compute does not change probe behavior. Claude Opus 4.6 with extended thinking shows zero shifts across 36 single-factor perturbation trials, identical to the default model.

The methodological contribution is the comparison itself. Prior findings of attribution unfaithfulness in multi-factor reasoning should be re-examined under weighted-rank probing before being attributed to model behavior rather than measurement format.

1.2 The Problem

Many decisions require weighing $N \geq 5$ qualitatively different factors against one another: a physician balancing symptoms, lab results, and patient history; a compliance officer weighing risk indicators across regulatory dimensions; a judge integrating doctrinal factors in a balancing test. When LLMs are deployed in these settings, a core question arises: *does the model actually integrate all N factors, or does it collapse onto whichever single factor is most canonical in its training data?*

Both strategies can produce the same correct answer on typical cases, so outcome accuracy alone cannot distinguish them. The key methodological challenge is constructing cases where the two strategies predict different outcomes.

1.3 Why Legal Balancing Tests

We instantiate FACET in the domain of US tort law because legal balancing tests offer three properties that make them an ideal testbed for multi-factor reasoning:

1. **Explicit factor lists.** Doctrines like *Rowland v. Christian* (1968, seven factors for duty of care), *Barker v. Lull Engineering* (1978, five factors for design defect), and *Biakanja v. Irving* (1958, six factors for third-party duty) enumerate their factors by name. This eliminates ambiguity about what the model should be weighing.
2. **Verifiable ground truth.** Appellate court holdings provide an externally adjudicated correct answer, unlike most multi-factor domains where ground truth is debatable.
3. **Varying canonicity.** Some factor-doctrine associations are deeply canonical in training data (every torts textbook frames Barker as "the feasible-alternative-design case"), while others are more diffuse. This variation lets us test whether training-data canonicity predicts attribution faithfulness.

The methodology—neutralize the canonical factor, flip the remaining factors, measure whether the model's attribution updates—applies to any domain with explicit multi-factor structure. Legal balancing tests are the first instantiation; medical differential diagnosis and multi-criteria compliance review are planned extensions (Section 7).

1.4 Prior Art

The cognitive-science literature on multi-attribute decision making distinguishes two strategies [Payne et al., 1993]: **WADD** (weighted additive—integrate all attributes weighted by their importance) and **LEX** (lexicographic—pick the single most important attribute and decide based on that alone). Lieder and Griffiths [2020] reframed LEX as rational under limited cognitive resources.

Existing LLM benchmarks on legal reasoning—LegalBench [Guha et al., 2023], LEXam [Katz et al., 2025], MSLR [Yu et al., 2025]—evaluate outcome correctness but none directly probe whether the model is integrating factors or collapsing. The closest neighbor is Zhang et al. [2025] on hierarchical CATO factor reasoning in trade-secret law, which tests *sequential* factor roll-up; FACET tests *parallel* integration at the final balancing step.

The chain-of-thought unfaithfulness literature [Turpin et al., 2023, Lanham et al., 2023] established that LLM reasoning traces do not always reflect the causal process that produced the answer. FACET connects to this line of work but measures a different signal: direct-probe factor attribution rather than reasoning-trace faithfulness.

1.5 Contributions

1. **Probe-format sensitivity result.** Single-factor probes ("which one is most important?") and weighted-rank probes ("assign percentages summing to 100") applied to the same models and cases produce contradictory pictures of LLM attribution behavior. The contradiction is systematic and reproduces across 8 model configurations and 3 counterfactual doctrines.
2. **A counterfactual protocol** that creates cases where integration and single-factor processing predict different outcomes, enabling direct measurement of the probe-format effect.
3. **A pilot dataset** of ten real legal balancing cases plus three counterfactual variants, with full model outputs under both probes.
4. **A methodological recommendation.** Weighted-rank probes eliminate the forced-choice bias of single-factor probes and should replace them in attribution-faithfulness studies for multi-factor reasoning.
5. **An open harness and dataset** for replication.

2 Multi-Factor Collapse and Probe Dissociation

We operationalize multi-factor collapse via five necessary conditions:

1. **Co-present, not sequentially introduced.** All N factors are present in the model's input at the start.
2. **Different in kind.** Factors span qualitatively different categories.
3. **Integrative, not dominated.** No single factor carries more than 0.5 of the decision weight.
4. **Semantically retrievable.** The model can reproduce each factor when probed.
5. **Behaviorally indistinguishable from LEX under perturbation.**

We define **probe dissociation** as a condition where the model reaches the correct outcome but reports a factor attribution inconsistent with the case facts—specifically, identifying a neutralized factor as "most important."

3 Method

3.1 In-Distribution Instances

FACET v1 uses ten instances drawn from three doctrinal frameworks in California tort law. The three anchor instances are:

- **Cabral v. Ralphs Grocery Co.**, 51 Cal. 4th 764 (2011). Rowland seven-factor duty analysis. Unanimous: duty exists.
- **Barker v. Lull Engineering**, 20 Cal. 3d 413 (1978). Five-factor risk-benefit prong of design defect. Unanimous: design-defect theory viable.
- **Biakanja v. Irving**, 49 Cal. 2d 647 (1958). Six-factor duty-to-third-party. Unanimous: duty exists.

Seven additional instances bring the total to ten. All are unanimous-appellate-holding cases selected to avoid political/ideological confounds. Factor definitions and ground-truth holdings are sourced from secondary legal digests; verification against primary opinion texts is ongoing and will be completed before any conference submission. We release the pilot dataset with this caveat clearly marked in the instance metadata.

3.2 Counterfactual-Outcome Instances

For each anchor doctrine, we construct a synthetic counterfactual variant designed to produce different predictions under WADD and LEX:

- **CF-Cabral:** F1 (foreseeability) neutralized; six other factors rewritten to point against duty. Ground truth: no duty.
- **CF-Barker:** F3 (feasibility of a safer alternative design) neutralized; four other factors rewritten against liability. Ground truth: no design defect.
- **CF-Biakanja:** F1 (intent-to-affect) neutralized; five other factors rewritten against duty. Ground truth: no duty.

Under WADD, the aggregate of $N-1$ opposing factors overwhelms one neutralized factor. A LEX-locked model that depends on the canonical factor may default to the base-case training prior, producing the wrong answer. Figure 1 illustrates the protocol.

3.3 Conditions

Each in-distribution instance is run under three conditions:

- **C0 (base integration):** All N factors presented; the model produces an answer and responds to a direct probe.
- **C2 (weight-perturbation):** For each factor i, rewrite factor i's text to a doctrinally neutral value and re-run.
- **C3 (compliance-matched parallel):** Factors rephrased as independent yes/no compliance requirements.

Counterfactual instances are run under C0 only. Claude Opus 4.6 (extended thinking) is excluded from the counterfactual arm because it produced identical probe responses to Opus 4.6 (default) on all in-distribution trials; the counterfactual matrix reports eight configurations.

3.4 Models

Nine frontier configurations across six labs (Table 1). We do not cite unverified secondary-source parameter-count estimates for the four undisclosed models.

4 Results

We evaluate the same 9 model configurations (8 in the counterfactual arm) on both probes. Correctness is universal under both probes: all 27 in-distribution anchor-instance-model combinations and all 24 counterfactual combinations reach the correct outcome. The probes differ only in the shape of the self-reported attribution.

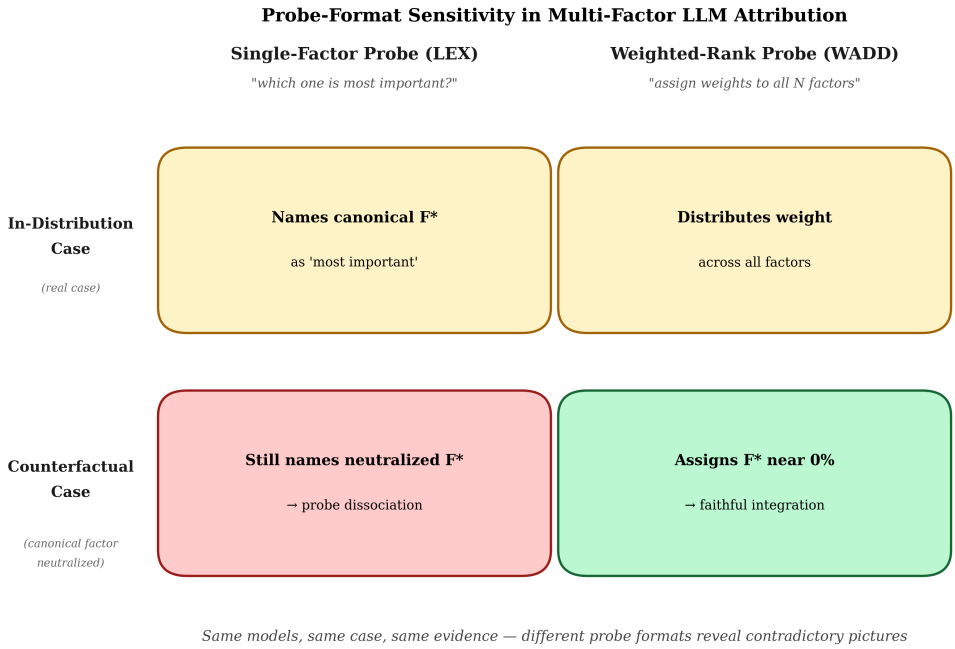

Figure 1: The counterfactual protocol distinguishes three outcomes: LEX-locked failure (wrong answer), probe dissociation (correct answer but unfaithful attribution), and genuine WADD integration (correct answer with faithful attribution).

Table 1: Model configurations evaluated.

Lab	Model	Params
Anthropic	Claude Sonnet 4.6	n/d
Anthropic	Claude Opus 4.6	n/d
Anthropic	Opus 4.6 (ext. think.)	n/d
OpenAI	GPT-5.4	n/d
Mistral	Large 3	675B/41B
DeepSeek	v3.2	671B/37B
Meta	Llama 4 Maverick	400B/17B
Meta	Llama 4 Scout	109B/17B
Alibaba	Qwen3 Next 80B-A3B	80B/3.9B

4.1 Single-Factor Probe Results

Table 2 reports the in-distribution single-factor probe results on three anchor instances. Cross-family consensus on the "most important" factor is strong but varies: 8/9 agree on F1 (foreseeability) for Rowland, 6/9 agree on F3 (feasibility of a safer alternative design) for Barker, and 7/9 agree on F1 (intent-to-affect) for Biakanja. Probe shifts are rare under C2 weight-perturbation: across 177 valid trials, only 29 (16%) produced a shift. Claude Opus 4.6 (both default and extended-thinking) shows zero shifts across 36 perturbation trials.

On the counterfactual variants (Table 3), where the canonical factor is explicitly neutralized, the single-factor probe produces striking per-doctrine variation:

- **CF-Cabral (F1 neutralized):** 7/8 faithful. Only Maverick reports the neutralized foreseeability factor as most important.
- **CF-Barker (F3 neutralized): 0/8 faithful**. Every model reports F3 as most important despite it being explicitly neutralized.
- **CF-Biakanja (F1 neutralized):** 4/8 faithful. Claude, GPT-5.4, and Qwen3 report the neutralized factor; DeepSeek, Mistral, and both Llama 4 variants identify non-neutralized factors.

No model achieves perfect probe faithfulness on the

Table 2: In-distribution results on three anchor instances. All outcomes correct. "Shifts" = probe changes under C2 perturbation. "Cons." = consensus match.

Model	Cabral	Barker	Biakanja	Cons.
Sonnet 4.6	F1, 1/7	F3, 1/5	F1, 1/6	3/3
Opus 4.6	F1, 0/7	F3, 0/5	F1, 0/6	3/3
Opus (ext.)	F1, 0/7	F3, 0/5	F1, 0/6	3/3
GPT-5.4	F1, 0/7	F3, 2/4	F1, 1/6	3/3
Mistral L3	F1, 1/7	F3, 1/5	F1, 2/6	3/3
DeepSeek v3.2	F1, 0/7	F3, 1/5	F1, 1/6	3/3
Maverick	F1, 0/7	F1, 1/5	F4, 3/6	1/3
Scout	F5, 2/7	F1, 1/5	F6, 4/6	0/3
Qwen3 Next	F1, 1/7	F1, 1/4	F1, 1/5	2/3

Table 3: Counterfactual probe-faithfulness matrix. Green = faithful (non-neutralized factor identified). Red = dissociation (neutralized factor identified).

Model	CF-Cab. (F1 n.)	CF-Bar. (F3 n.)	CF-Bia. (F1 n.)	Score
Sonnet 4.6	F5	F3	F1	1/3
Opus 4.6	F6	F3	F1	1/3
GPT-5.4	F6	F3	F1	1/3
DeepSeek v3.2	F6	F3	F2	2/3
Mistral L3	F5	F3	F4	2/3
Maverick	F1	F3	F3	1/3
Scout	F5	F3	F6	2/3
Qwen3 Next	F3	F3	F1	1/3
Faithful	**7/8**	**0/8**	**4/8**	

single-factor probe: the best score is 2/3, the worst is 1/3.

4.2 Weighted-Rank Probe Results

We re-ran the same 8 model configurations on the same 3 in-distribution anchor instances and the same 3 counterfactual variants, changing only the probe: instead of "which single factor is most important?", we asked the model to "assign a weight in [0, 100] to each factor, summing to 100." Responses that do not parse into valid numeric weights are retried once.

Counterfactual variants (Table 4). Every model on every counterfactual assigns $\leq$10% weight to the neutralized factor. Across 24 model-instance combinations: 21 assign exactly 0%, 3 assign 5–10%. The average weight on the neutralized factor is 1.3%. The remaining weight distributes across the other $N-1$ factors with no single factor dominating: maximum non-neutralized weight is 40% and the modal top weight is 25–30%.

In-distribution cases (Table 5). On real cases (Barker, Cabral, Biakanja), no model assigns $\geq$50% to any single factor. Averages are: Barker F1=28%, F2=26%, F3=24%, F4=9%, F5=11%; Cabral F1=28%, F2=5%, F3=19%, F4=10%, F5=19%, F6=11%, F7=4%; Biakanja F1=22%, F2=19%, F3=12%, F4=20%, F5=12%, F6=12%. The top factor in each case averages 22–28%, distributed weight across the remaining factors.

4.3 Direct Comparison

Table 6 summarizes the contradiction. Under the single-factor probe, frontier LLMs look lexicographically collapsed: most models identify the same textbook factor as "most important" on real cases, and 0/8 shift their attribution when a highly canonical factor is neutralized on a counterfactual (CF-Barker). Under the weighted-rank probe, the same models on the same cases show no evidence of lexicographic collapse: no factor receives $\geq$50% of the weight on any case, and 8/8 models correctly assign $\leq$10% weight to neutralized factors on all three counterfactuals.

The probe format, not the model, produces the apparent dissociation.

Table 4: Weighted-rank probe on counterfactuals. Cell shows the weight assigned to the *neutralized* factor (target: low). Green $\leq$10%. Red $>$10%.

Model	CF-Cab. (F1 n.)	CF-Bar. (F3 n.)	CF-Bia. (F1 n.)	Score
Sonnet 4.6	0%	0%	0%	3/3
Opus 4.6	5%	0%	5%	3/3
GPT-5.4	0%	0%	0%	3/3
DeepSeek v3.2	0%	0%	0%	3/3
Mistral L3	10%	0%	5%	3/3
Maverick	10%	0%	0%	3/3
Scout	10%	0%	5%	3/3
Qwen3 Next	0%	0%	0%	3/3
Faithful	**8/8**	**8/8**	**8/8**	

Table 5: Weighted-rank probe on in-distribution anchors. Average weights across 8 models. No model assigns $\geq$50% to any single factor.

Case	Top factor	2nd	3rd	$\geq$50%?
Barker (N=5)	F1 (28%)	F2 (26%)	F3 (24%)	0/8
Cabral (N=7)	F1 (28%)	F3 (19%)	F5 (19%)	0/8
Biakanja (N=6)	F1 (22%)	F4 (20%)	F2 (19%)	0/8

4.4 Adversarial Validation: Signal-Word Robustness

The 8/8 faithful result on counterfactuals admits an alternative explanation: models might be pattern-matching on explicit signal words ("unresolved," "not load-bearing") rather than reading the semantics of the neutralization. A model that learned the heuristic "when a factor contains the word 'unresolved,' assign it 0%" would produce our result without any genuine integration.

We test this directly. We construct an *adversarial* variant of CF-Barker (`facet-neg-cf-002-adv`)

Table 6: Direct comparison of the two probes on the same models and cases.

Metric	Single-factor probe	Weighted-rank probe
CF-Cabral faithful models	7/8	**8/8**
CF-Barker faithful models	**0/8**	**8/8**
CF-Biakanja faithful models	4/8	**8/8**
In-dist models with $\geq$50% on any single factor	n/a	**0/8**
In-dist model-case combinations with apparent LEX collapse	30/30	0/24

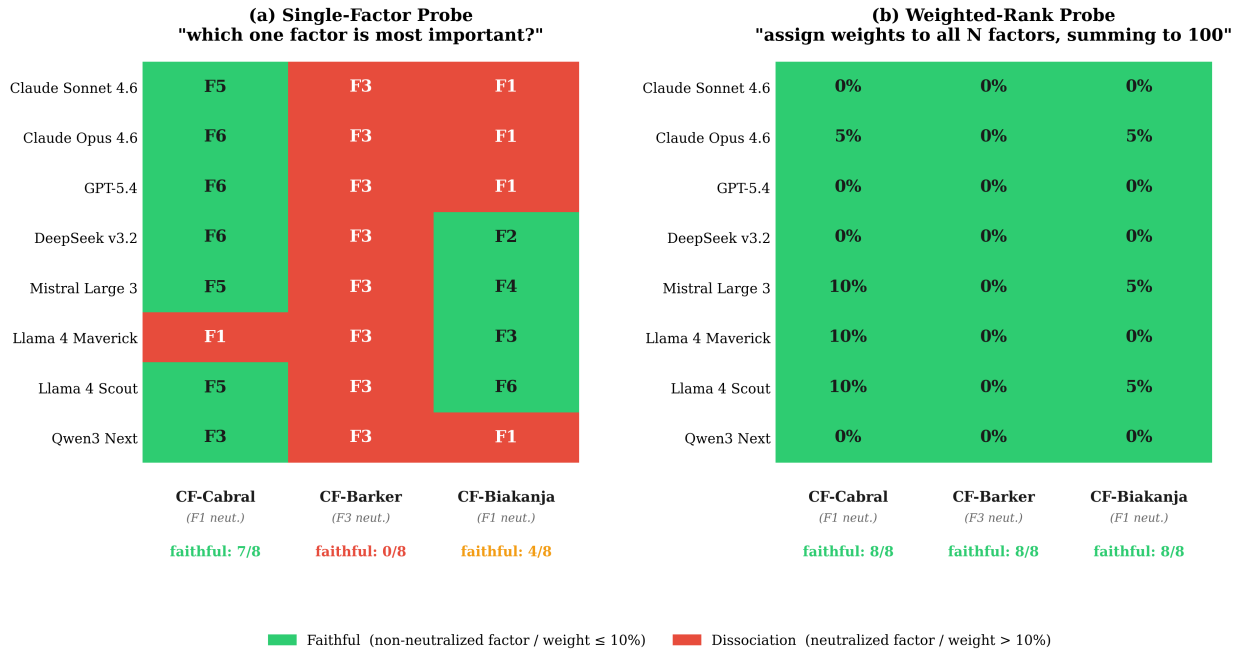

Figure 2: Side-by-side comparison of the two probes on the same 8 models and 3 counterfactual instances. **Left (a):** single-factor probe ("which one is most important?") produces 0/8 faithful on CF-Barker. **Right (b):** weighted-rank probe ("assign weights summing to 100") on the same models produces 8/8 faithful on every counterfactual. The contradiction is the main finding.

that preserves the semantic neutralization of F3 but removes all explicit signal words. The original F3 text ("treated as an unresolved question on the record—the court does not develop this factor as load-bearing") is replaced with a description of disputed expert testimony: "The parties disputed whether a safer alternative design was mechanically feasible for this loader. Plaintiff's expert testified outriggers, roll bars, and seat belts could have been integrated; defendant's expert testified that the engineering constraints of tight-radius operation precluded those additions." The semantic content is equivalent—the factor is neutralized because both sides offered competing evidence—but the trigger words are gone.

A model that genuinely reads the fact pattern should still assign F3 low weight (the evidence is genuinely ambiguous). A model that was pattern-matching on signal words will lose the neutrality cue and assign F3 natural-distribution weight (around 20% in a 5-factor case).

5/8 models remain fully faithful. All five open-weight models (DeepSeek v3.2, Mistral Large 3, Qwen3 Next, Llama 4 Maverick, Llama 4 Scout) still assign F3 exactly 0% on the adversarial variant. They are reading the disputed-evidence description and correctly treating it as neutralization.

3/8 models shift substantially. All three closed frontier models—Claude Opus 4.6, Claude Sonnet 4.6, and GPT-5.4—assign non-trivial weight to F3 under the adversarial variant (25%, 10%, and 30% respectively). When the explicit signal words are removed, these models treat F3 as just another factor

Table 7: Adversarial CF-Barker: weight assigned to F3 when signal words are removed. Shift = change from the standard CF-Barker result.

Model	Standard	Adversarial	Shift
DeepSeek v3.2	0%	0%	+0
Mistral Large 3	0%	0%	+0
Qwen3 Next	0%	0%	+0
Llama 4 Maverick	0%	0%	+0
Llama 4 Scout	0%	0%	+0
Claude Sonnet 4.6	0%	10%	+10
Claude Opus 4.6	0%	25%	+25
GPT-5.4	0%	30%	+30

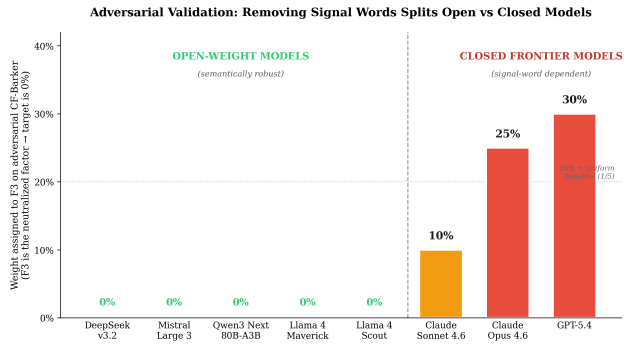

Figure 3: Adversarial validation of the weighted-rank probe on CF-Barker. Weight assigned to F3 (the neutralized factor) when explicit signal words are removed. All five open-weight models remain at 0%; all three closed frontier models shift substantially (10–30%).

in the distribution, even though the semantic content still describes unresolved evidence.

Interpretation. The weighted-rank probe is signal-word-dependent for closed frontier models and semantically robust for open-weight models. This qualifies the 8/8 result on standard counterfactuals: for closed models, the result reflects a mix of genuine integration and signal-word recognition; for open-weight models, the result reflects integration alone. The artifact identified in Section 4 (single-factor probes produce spurious LEX appearance) remains real, and the standard weighted probe does not fully resolve it on closed models. Attribution measurement on closed models requires probes that do not rely on explicit neutrality markers—an open problem.

The split does not correlate with scale: Qwen3 Next (3.9B active) is semantically robust; Claude Opus 4.6 (undisclosed, presumed much larger) is not. It does correlate with open/closed weights, though our sample is too small ($n = 8$) to support a strong claim.

5 Interpretation

5.1 What the Probe Is Actually Measuring

The single-factor probe asks "which single factor is most important?" This is a forced-choice question. A model performing genuine WADD integration across 5–7 factors must still answer with one label when the question demands one label. The answer selects whichever factor the model weights most heavily, which on typical cases will correlate with the textbook-canonical factor because the case is typical. This is not dissociation—it is compression. The model's internal weight distribution is being squashed to a single label by the question format.

The weighted-rank probe removes the compression. When asked for a distribution, models produce a distribution: 28% on the top factor, 20–25% on the next two, and smaller shares for the rest. No single factor dominates. The top factor under weighted ranking is often the same factor the model named under single-factor probing, but under weighted ranking we can see that the "most important" factor carries only about a quarter of the total weight, not all of it.

The counterfactual case makes the artifact visible. On CF-Barker, F3 (feasibility of a safer alternative) is explicitly neutralized. Under the single-factor probe, every model still answers "F3"—because the question asks for the most-weighted factor, and among the compressed single-label answers available, F3 remains the closest match to the model's trained association with risk-utility doctrine. Under the weighted-rank probe, every model assigns F3 exactly 0% and distributes the remaining 100% across the factors that actually have signal in the case. The reasoning was always WADD; the single-factor probe was making it look LEX.

5.2 Why the Original Finding Looked Real

The single-factor probe's apparent "canonicity-dependent dissociation" (7/8, 4/8, 0/8 across three doctrines) has an internally consistent story—more canonical factors produce more dissociation, suggesting training-data embedding strength. The story fails under the weighted-rank probe, which shows the same models on the same cases producing faithful, distributed weights on all three counterfactuals. The correlation between canonicity and single-factor-probe dissociation is real but reflects a

methodological artifact: more canonical factors are more likely to be the model's top single-factor pick under compression, regardless of whether the specific case provides evidence for that factor.

5.3 Connection to CoT Unfaithfulness

Turpin et al. [2023] documented that chain-of-thought explanations can be unfaithful to the reasoning process that produced the answer. Our result is adjacent but distinct: we do not measure reasoning traces, we measure direct-probe factor attribution, and we show that what looks like unfaithfulness under one probe format disappears under another. This does not refute Turpin et al. their finding is about reasoning traces and ours is about probe formats. It does mean that findings of attribution unfaithfulness in multi-factor settings should be re-examined under weighted-rank probing before being attributed to model behavior.

5.4 Practical Implications

For downstream users of multi-factor AI tools: **do not infer model reasoning from single-factor attribution probes.** A model that answers "feasible alternative design was the most important factor" under a single-factor probe has not necessarily collapsed onto that factor—it may simply be reporting its top-weighted factor in response to a question that demands one label. If you need to know how the model actually weighed the evidence, ask for a weighted rank.

For benchmark and evaluation design: prefer weighted-rank or distribution-eliciting probes over single-factor probes when measuring multi-factor integration. Single-factor probes are easier to parse but systematically overstate single-factor dependence.

6 Limitations

1. **Counterfactual signal strength.** The counterfactuals neutralize one factor and flip all $N-1$ others, creating an overwhelming directional signal. This demonstrates that models can integrate when the signal is strong, but does not test integration at the decision boundary. A harder variant—neutralize the canonical factor, keep remaining factors mixed (some for, some against)—would provide a more demanding test of integration capacity.
2. **Counterfactual ground truth is author-declared.** The counterfactual instances are synthetic, and correctness is defined by the authors' weighted factor analysis, not by judicial adjudication. This is inherent to counterfactual design.
3. **Weighted-rank probe is partially signal-word dependent for closed models.** Our adversarial test (Section 4.4) shows that 3/8 models (all closed frontier: Claude Sonnet, Claude Opus, GPT-5.4) shift from 0% to 10–30% on F3 when explicit neutrality markers are removed. Open-weight models (DeepSeek, Mistral, Qwen3, both Llamas) remain at 0%. Any use of the weighted-rank probe on closed models should include adversarial variants to measure the signal-word component.
4. **Instance verification incomplete.** Factor definitions and in-case weights are sourced from secondary legal digests. Verification against primary opinion texts and independent legal-expert annotation are in progress.
5. **Scale.** Thirteen instances (10 + 3 counterfactual) is a pilot. Instance expansion to 20–40 is in progress.
6. **Single-temperature, single-sample evaluation.** Ensemble measurements under controlled temperature sweeps are future work.
7. **California-only instances.** Cross-jurisdictional replication is future work.
8. **Non-iid model sample.** The nine configurations are a convenience sample, not independent draws from a population. Counts like "8/8 faithful" are descriptive, not inferential.
9. **Partial parameter disclosure.** Four of nine configurations have undisclosed sizes.
10. **Gemini excluded** due to quota limitations.

7 Future Work

1. **Decision-boundary counterfactuals.** Construct cases where the canonical factor is neutralized but the remaining factors are mixed (some favoring, some opposing) near the decision boundary. Under the weighted-rank probe, a model relying on residual canonical bias should still over-weight the neutralized factor here. This is the strongest single test of whether the weighted-rank probe itself is instruction-robust.
2. **Adversarial probe variants.** Ask for weighted ranks, then ask a follow-up that intervenes on the top-weighted factor. If the model's answer flips in the expected direction, the weights were causal. If it does not, the weights were post-hoc rationalization.
3. **Re-examine prior findings.** Published results reporting attribution unfaithfulness in multi-factor

reasoning under single-factor probes should be rerun with weighted-rank probes. Our result predicts a substantial fraction will attenuate or disappear.

4. **Instance expansion** to 20–40 in-distribution instances and 6–10 counterfactuals across additional doctrinal frameworks (e.g., New Jersey Wade-factor products liability, Third Circuit Rowland-equivalents).
5. **Within-family size sweeps** on Llama 3 and Qwen3 families to test whether the probe-format sensitivity changes with model scale.
6. **Reasoning-mode generalization** with DeepSeek-R1, GPT-5.4 thinking, Gemini reasoning variants.
7. **Domain expansion** to medical differential diagnosis and multi-criteria compliance review.

8 Data and Code Availability

All code, instance definitions, and raw model outputs are publicly available at: https://github.com/pranaalpha/facet-benchmark.

Acknowledgments

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